

Srey Jaiswal

+91 6371714883 | shreyjaiswal2005@gmail.com | LinkedIn | GitHub | Certificates

EDUCATION

SRM Institute of Science and Technology

B.Tech in Electronics Engineering (VLSI Design & Technology)

CGPA: 9.08/10

Chennai, India

2024 - 2028 (Expected)

Kenbridge School

CBSE Board: 76.2%

Kota, India

2022-2024

EXPERIENCE

D.D. Iron & Steel (P) Ltd.

Jun - Jul 2026

Electrical Systems Intern

- Assisted in inspection and understanding of induction-furnace electrical systems, including transformers, control panels, induction coils, cooling circuits, and electrical measurements using multimeters and clamp meters.
- Led a four-member team on electrical quality-assurance measurements using multimeters and clamp meters.

Team 1.618, SRM IST

2024 - Present

Electrical Team Member

- Contributed to HV/LV electrical integration, 13S14P accumulator and BMS systems, BSPD/shutdown circuitry, pre-charge, contactors, protection, and vehicle-level testing.

The Faith, Rourkela

Jun - Jul 2026

Social Development Coordinator Intern

- Organized food and clothing distribution drives and assisted people with disabilities during community and religious outreach activities.

PROJECTS

FPGA-Based INT8 Neural-Network Accelerator using a 4x4 Systolic Array

2025-2026

Tech used: SystemVerilog | Xilinx Vivado | PYNQ-Z2 | AXI | FPGA | Python | NumPy

- Designed and implemented a 4x4 output-stationary systolic-array accelerator with 16 pipelined MAC processing elements, supporting signed INT8 activations and weights with 32-bit accumulation.
- Developed shared-BRAM-based data exchange, local tile buffering, data scheduling, address generation, and FSM-based control for automated tiled matrix-multiplication execution.
- Integrated the accelerator with the ARM Cortex-A9 processing system on PYNQ-Z2 using AXI4-Lite control and AXI-accessible shared BRAM for hardware-software co-design.
- Extended the matrix engine for quantized neural-network inference using bias addition, ReLU activation, INT8 requantization/saturation, and a tiled 16-input x 16-output dense-layer demonstration.
- Implemented the accelerator at 100 MHz and validated functionality using self-checking SystemVerilog testbenches and Python/NumPy golden models, while evaluating latency, throughput, and FPGA resource utilization.

5-Stage Pipelined RV32I RISC-V Processor on PYNQ-Z2

2025-2026

Tech used: Verilog | Xilinx Vivado | ModelSim | PYNQ-Z2 | RISC-V

- Designed and implemented a 32-bit RV32I processor using a 5-stage IF-ID-EX-MEM-WB pipeline with 4 pipeline registers and a 32 x 32-bit register file.
- Implemented EX/MEM and MEM/WB forwarding, 1-cycle load-use hazard stalls, and pipeline flushing with NOP insertion for 6 branch types, JAL, and JALR.
- Supported arithmetic, logic, shift, comparison, load/store, immediate, branch, jump, LUI, and AUIPC instructions with signed/unsigned byte, half-word, and word memory operations.
- Deployed the processor on the PYNQ-Z2 FPGA and validated correct register and memory outputs using a 42-instruction test program after synthesis, implementation, timing analysis, and bitstream generation in Vivado.

Formula Student Hybrid Vehicle Electrical System & BSPD

2024-2025

Tech used: Battery Systems | BMS | HV/LV Systems | Electrical Safety | KiCad

- Contributed to HV/LV electrical design and integration for a Formula Student hybrid vehicle.
- Worked on a 13S14P Li-ion accumulator (182 cells, 48.1 V nominal, 54.6 V max) with pre-charge, contactors, protection, monitoring, and isolation.
- Integrated the Orion Jr. 2 CAN BMS and supported BSPD/shutdown-system testing and vehicle-level electrical validation.

TECHNICAL SKILLS & INTERESTS

HDL & RTL:	Verilog, SystemVerilog, RTL Design, Digital Logic Design
FPGA & EDA:	Xilinx Vivado, ModelSim, PYNQ-Z2
Computer Architecture:	RISC-V, RV32I, Pipelined Processors, Systolic Arrays
Hardware Interfaces:	AXI4-Lite, BRAM, FPGA-ARM Hardware/Software Integration
Programming & OS:	C, Python, NumPy, Assembly, Linux
Embedded Systems:	Microcontroller Interfacing, Hardware-Software Integration
Hardware Design:	PCB Design, KiCad
Electrical Systems:	Battery Management Systems, HV/LV Systems, Electrical Safety

LANGUAGES

Spoken & Natural:	English (Professional/Academic), Hindi (Native), Odia (Native), Tamil (Conversational)
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ACHIEVEMENTS

- 1st Prize | Konnect Case Quest | Won 1st prize in the Case Quest competition at SRM IST, demonstrating case analysis, strategic thinking, and presentation skills (2026)
- 1st Prize | Technical Quiz, Mini Colloquium on Semiconductor Manufacturing Technologies | Secured 1st prize in the inter-college technical quiz at SRM IST (2026)
- City Topper | JEE Main, Rourkela | Recognized as JEE Main City Topper in Rourkela (2024)

VOLUNTEERING

- Technical Volunteer | VEGA Processor Workshop, SRM IST | Assisted with software setup, troubleshooting, and hands-on sessions using VEGA/ARIES RISC-V boards (2026)
- Volunteer | Dibya Dham Trust | Supported underprivileged communities and elderly residents through old-age home assistance, city cleanliness drives, and community service activities (2020-Present)